Fred (Zeyuan) Xu

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Summary

Computer Science Ph.D. student at UCLA researching probabilistic generative modeling (flow matching, diffusion), graph learning, and uncertainty quantification, with applications to finance and science. Six first- or co-first-author papers at **NeurIPS (4)**, **KDD**, and **ACL**; reviewer for NeurIPS, ICML, and ICLR. Industry research at Block (two AI research internships), Hitachi, and Amazon, applying this work to financial fraud detection, customer risk modeling, transaction foundation models, and recommendation.

Education

- **University of California, Los Angeles** — Los Angeles, CA
  Ph.D. in Computer Science — *Sep 2022 – Present*
- **University of California, Los Angeles** — Los Angeles, CA
  B.S. in Computer Science and Engineering; Minor in Mathematics — *Sep 2013 – Mar 2018*

Professional Experience

- **Block, Inc.** — Oakland, CA
  AI Research Intern — *Jan 2026 – Sep 2026*
  - **Function-Space Flow Matching**: Proposed kernel Functional Flow Matching (kFFM), replacing independent prior–data pairing with entropic optimal transport under a kernel-induced cost, with well-posedness and discretization-invariance guarantees; improved distributional matching over FFM, diffusion, and OT baselines on time-series and PDE benchmarks. First-author paper at **NeurIPS 2026**.
  - **Graph Anomaly Detection for Fraud**: Developed EB-GAD, a training-free, label-free anomaly detector that fits a graph prior by empirical Bayes and scores nodes by closed-form finite-horizon control energy; best or tied-best AUROC on 9 of 11 benchmarks, including four financial fraud networks of up to 3.7M nodes, with margins of up to 21.7 points. First-author paper at **NeurIPS 2026**.
  - **Uncertainty Quantification for GNNs**: Proposed DSS-GNN, which expands node embeddings on graph-spectral and Wiener-chaos axes so a single representation supports calibrated prediction, out-of-distribution detection, and distribution-shift robustness. First-author paper at **NeurIPS 2026**.
  - **Graph Generation**: Co-developed FlowPhon, which generates dense graphs via flow matching on graphon operators; second-author paper under review.
  - **Transaction Foundation Model**: Built a GPT-style foundation model over customer transaction event sequences to model Cash App balance changes.
- **Handshake AI, Inc.** — Remote
  STEM Specialist — *Aug 2025 – May 2026*
  - **LLM Post-Training Data**: Authored and solved advanced STEM problems and evaluated LLM-generated code to support alignment and improve the generation quality of LLM agents.
- **Block, Inc.** — Oakland, CA
  AI Research Intern — *Jan 2025 – Sep 2025*
  - **Uncertainty Estimation on Graphs**: Developed a graph neural network grounded in stochastic partial differential equations for uncertainty estimation on graph data; first-author paper at **NeurIPS 2025**.
  - **Customer Risk Modeling**: Implemented a probabilistic machine learning model with Monte Carlo simulation to estimate customer default risk and account lifetime value.
  - **Time-Series Generation**: Created a distributional diffusion model for time series built on path signatures.
- **Hitachi America, Ltd.** — Santa Clara, CA
  Research Scientist Intern — *Jun 2024 – Sep 2024*
  - **Sensorless Cargo Monitoring**: Engineered features and built regression and time-series models on global cargo-container temperature and humidity data to estimate shipment operational risk, improving on the baseline model by 30%.
  - **Financial Risk Simulation**: Ran Monte Carlo simulations over financial indicators and corporate portfolios, combined with time-series analysis, to estimate corporate financial risk under currency-rate and market shifts.
- **Amazon.com, Inc.** — New York, NY
  Applied Scientist Intern — *Jun 2022 – Sep 2022*
  - **Audience Embedding for Recommendation**: Implemented a deep-learning bipartite graph embedding model for audience representation in recommendation and user-group behavior prediction; proposed fuzzy-set embeddings to support logical queries, outperforming the baseline by at least 30%.

- **NortonLifeLock Inc.** Culver City, CA
  Software Engineer *Jul 2018 – May 2021*
  - **Privacy Monitor (iOS)**: Developed the privacy monitor feature and client kit in Swift for the Norton Secure VPN iOS app, increasing customer interest in the privacy product by 45%.
  - **Norton Secure VPN (Android)**: Maintained and extended the Android app (10M+ installs), shipping features including split tunneling and torrenting support.
  - **Norton Security UWP**: Developed, integrated, and unit-tested core C#, Angular, and C++/WinRT code for the Universal Windows Platform app; helped build the product threat model and implemented encryption, authentication, and file-integrity mechanisms.
  - **Secure File Transfer MVP**: Built an MVP across Android, iOS, and PWA (Xamarin, Angular) for secure file transfer using TLS, RSA, AES-256, and SHA-256, with key exchange and management via Azure Key Vault.
  - **White Glove Service MVP**: Developed a Chrome extension for a service that opts users out of data-broker websites to protect their privacy.

Publications

NeurIPS'26 Improving Function Space Flow Matching with Kernel Optimal Transport. *First author*

NeurIPS'26 Graph Anomaly Detection as Finite-Horizon Control: Training-Free Scoring via Empirical Bayes. *First author*

NeurIPS'26 A Unified Uncertainty Representation for Graph Neural Networks via Doubly-Spectral Stochastic Expansion. *First author*

KDD'26 FD-Bench: A Modular and Fair Benchmark for Data-driven Fluid Simulation. *Co-first author. arXiv:2505.20349*

KDD'26 TIDAL: A Temporal Causal Diffusion Framework for Visualizing Knee Osteoarthritis Treatment Outcomes. *Second author*

ACL'26 Training Recipes for Agentic Reinforcement Learning in LLMs: A Survey. *Co-author*

NeurIPS'25 Uncertainty Estimation on Graphs with Structure Informed Stochastic Partial Differential Equations. *First author. arXiv:2506.06907*

ACL'24 FUSE: Measure-Theoretic Compact Fuzzy Set Representation for Taxonomy Expansion. *First author. arXiv:2506.08409*

Manuscript Neural Hybrid Residual XRO Models for ENSO Forecasting. *First author*

Submitted FlowPhon: Graph Generation via Flow Matching on Graphon Operators. *Second author*

Skills

- **Programming**: Python, C++, SQL, Shell
- **ML & Data**: PyTorch, PyTorch Geometric, JAX, Hugging Face, scikit-learn, NumPy, pandas, Spark, AWS
- **LLMs**: Fine-tuning and evaluation; OpenAI, Claude, and Gemini APIs
- **Methods**: Flow matching, diffusion models, uncertainty quantification, graph learning, time series, scientific ML
- **Spoken Languages**: English, Chinese, German, Spanish

Teaching

- **University of California, Los Angeles** Los Angeles, CA
  Instructor and Teaching Assistant
  - **UCLA Summer Institute Instructor (Jul – Aug 2023)**: Taught **CS 97: Introduction to Data Science** to advanced high school students for college credit; rated **8.75/9** overall.
  - **Teaching Assistant**: CS 148 (Introduction to Data Science), CS 247 (Advanced Data Mining), CS 33 (Computer Organization), CS M51A (Logic Design of Digital Systems); average **8.4/9** overall rating.

Service & Talks

- **Reviewer**: NeurIPS, ICML, ICLR, WWW, CIKM; PLOS journals
- **Talks**: Presented machine learning papers and tutorials at the UCLA Data Mining study group, student clubs, and company knowledge-sharing meetings.